

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Global climate change - Set 2

Mark scheme

AQA spec 3.2.1.2; The physical environment; textbook pp. 89-107

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Raise river banks / build sea walls, flood barriers or wave screens (1) • Permeable urban surfaces (gravel, soil) instead of concrete / tarmac to slow runoff (1) • River flow management: upstream dams / low soil dams / afforestation / maintained flood plains to delay runoff (1) • Raised buildings on stilts or floating houses (1) • Pumping rainwater from land below river level (1) • Managed retreat - abandoning land where protection costs exceed value (eg Happisburgh) (1)	3
01.2	<i>1 mark for each point.</i> • Where the cost of flood prevention is greater than the value of what is protected, or defences would be ineffective (1) • Land is abandoned as water levels rise / coast erodes; defences used only where land is very important (1)	2
01.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Road surfaces deform as tar melts - need tarmac with a higher melting point (1) • Rail track expands and buckles above the temperature it was pre-stretched for (27 °C in UK) - relay with higher pre-stress (1) • Bridge damage from high river flows / debris; landslides on deforested slopes (1)	2

Question	Marking guidance	Marks
01.4	<i>1 mark for each point.</i> • CO₂ is captured from the flue gases (or carbon removed from the fuel before combustion) (1) • Transported by pipeline / ship / tanker and stored in depleted oil or gas fields, aquifers, or used for secondary oil recovery (1) • Limitation: developmental / expensive, uses energy (reducing efficiency), risk of leakage, limited storage sites (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 99-100, 105-108; AQA spec 3.2.1.2.	
02.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Control of greenhouse gases: energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture • Geoengineering: white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences • International agreements: Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation; carbon trading and taxes; comparison of per capita carbon footprints between countries • Evidence of limited success: global CO₂ concentrations still rising (about 280 ppm pre-industrial to over 410 ppm); temperatures up about 1 °C; emissions in some developed countries falling but rising in developing economies • Reasons for difficulty: fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop • Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause • Comparison with the Montreal Protocol's success for ozone • Judgement: partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise; tipping points make urgency greater	9
02.2	<i>1 mark.</i> • The total greenhouse gas / CO₂ emissions caused by an individual, organisation, product or country (often expressed per capita per year) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2.	

Question	Marking guidance	Marks
03.1	<i>1 mark for each point.</i> • 2020 has higher peak flow in the wet season (32 vs $22 \text{ m}^3 \text{ s}^{-1}$ in March) (1) • 2020 has much lower dry-season flow (2 vs $9 \text{ m}^3 \text{ s}^{-1}$ in July / August) (1) • Flow is much more variable / seasonal in 2020; the river almost dries up (1)	3
03.2	<i>1 mark for each point.</i> • In 1990 snow / ice stored wet-season precipitation and released it gradually as meltwater during the dry season, evening out the flow (1) • With less ice, precipitation now falls as rain / melts immediately and runs off quickly, increasing wet-season flow and flood risk (1) • Little stored ice remains to melt in the dry season so flow falls very low (1)	3
03.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Water shortages for irrigation / drinking / HEP in the dry season (1) • Flooding in the wet season (1) • Sudden floods from ice-lake outbursts (glacial lake outburst floods) (1)	2
03.4	<i>1 mark for each point.</i> • Surface meltwater flows down cracks to the base, lubricating the ice so it slides faster; if this exceeds the faster melting at the front, the front advances (1) • But if snowfall at the source has not increased, total volume falls; both retreating and extending glaciers can indicate warming (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	